

sorting speed in MB/s

2500

2000

1500

1000

500

0

Icelake, Intel Xeon Gold 6338, 2 GHz
378 GiB RAM

parallel, 4 threads
radsort, 1 thread
generic, 1 thread
bitmanip, 1 thread
swc, 1 thread

$\sigma = 256$ (one byte)
 $n_t = 4$ (key); 4 bytes val.
 $b = 512$

cacheline ->

L1 cache ->

L2 cache ->

L3 cache ->

RAM

$2^5 = 32$ B

$2^{10} = 1$ KiB

$2^{15} = 32$ KiB

$2^{20} = 1$ MiB

$2^{25} = 32$ MiB

$2^{30} = 1$ GiB

$2^{35} = 32$ GiB

array length [bytes]

△ larger is better

